



# LOST BOYS CYBER

THREAT INTELLIGENCE • MALWARE ANALYSIS • DETECTION ENGINEERING



## Direct-to-IP Malware C2 and DNS Bypass Detection Daily Update 2026-09-05

Threat Report

|                       |                 |
|-----------------------|-----------------|
| <b>Classification</b> | TLP:CLEAR       |
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| <b>Author</b>         | Lost Boys Cyber |
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TLP:CLEAR | Lost Boys Cyber | Direct-to-IP Malware C2 and DNS Bypass Detection Daily Update 2026-09-05 - Threat Report

Published: 2026-09-05 | TLP:CLEAR | Version: 1.0 | Author: Lost Boys Cyber

AI Generation: This report was generated with AI assistance and is provided for informational purposes only. All findings, IOCs, detection rules, hunting queries, attribution assessments, and recommendations must be independently reviewed and validated by a qualified security professional before operational use.

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# Direct-to-IP Malware C2 and DNS Bypass Detection Daily Update 2026-09-05

## 1. Daily Freshness Note

This 2026-09-05 daily-update edition creates a current-day GitHub-visible artifact for a still-relevant priority topic. Sources were rechecked during the automated run; stale claims were not promoted, and defender actions were refreshed for today's operational queue.

## 2. Executive Summary

This report was selected for the 2026-09-05 UTC Lost Boys Cyber daily coverage set because it presents actionable defender relevance, current source coverage, and clear opportunities for detection or threat hunting. The activity or trend should be reviewed by SOC, vulnerability management, identity, and sector-risk owners.

Primary exposure: Organizations relying heavily on DNS security controls without equally strong egress and proxy telemetry. The report does not assert victim compromise beyond public source claims; it converts available reporting into defensive priorities, hunt hypotheses, and operational controls.

## 3. Intelligence Requirements

| Question                     | Current Answer                                                                                            | Confidence |
|------------------------------|-----------------------------------------------------------------------------------------------------------|------------|
| What happened?               | Public reporting describes Direct-to-IP Malware C2 and DNS Bypass Detection Priorities.                   | High       |
| Who should care?             | Organizations relying heavily on DNS security controls without equally strong egress and proxy telemetry. | High       |
| What can defenders do today? | Review exposure, run the included hunts, and validate mitigations against recent telemetry.               | High       |

## 4. Source Review & Confidence

| Source                          | Contribution                                                                                                          | Confidence  |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------|-------------|
| Unit 42 direct-to-IP malware C2 | Analysis of 4 million dynamic-analysis reports found 45.32% of malware samples with C2 used direct-to-IP connections. | Medium-High |
| Unit 42 threat research index   | Context for current Unit 42 research and incident-response findings.                                                  | Medium-High |
| M-Trends 2026                   | Frontline incident data for attacker persistence and detection priorities.                                            | Medium-High |

Source depth was sufficient for a defensive threat report. Hard IOCs were not consistently available in collected public sources, so this report emphasizes TTP and telemetry-based analytics.

## 5. Threat Activity Overview

Reporting points to Multiple malware families and intrusion operators activity/trends involving Application-layer C2, Direct IP egress, Proxy bypass, Defense evasion against DNS-only monitoring. For defenders, the highest-value action is to convert these observations into exposure review and telemetry hunts rather than waiting for environment-specific IOCs.

## 6. Affected Sectors and Exposure

| Exposure Area                       | Why It Matters                                                                                    | Recommended Review                                                                  |
|-------------------------------------|---------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Identity and remote access          | Valid accounts reduce attacker friction and bypass many perimeter controls.                       | Review MFA fatigue, impossible travel, help-desk reset and risky sign-in patterns.  |
| Internet-facing applications        | Public services remain common initial access and implant placement points.                        | Confirm patch status and inspect web logs for unusual POST/URI/user-agent patterns. |
| Endpoint and developer workstations | Lures and developer workflows can execute scripts or malware locally.                             | Monitor Node.js, PowerShell, shell and archive execution from user-writable paths.  |
| Third-party and shared platforms    | Aviation and other operators depend on vendors and SaaS that can become blast-radius multipliers. | Review vendor access, SSO integrations, and contingency procedures.                 |

## 7. Tactics, Techniques, and Procedures

| Tactic / Phase      | Observed Technique                      | Evidence                                                                                        |
|---------------------|-----------------------------------------|-------------------------------------------------------------------------------------------------|
| Initial Access      | T1566 Phishing                          | Application-layer C2 described in source reporting or analyst synthesis.                        |
| Execution/C2/Impact | T1059 Command and Scripting Interpreter | Direct IP egress described in source reporting or analyst synthesis.                            |
| Execution/C2/Impact | T1071 Application Layer Protocol        | Proxy bypass described in source reporting or analyst synthesis.                                |
| Execution/C2/Impact | T1078 Valid Accounts                    | Defense evasion against DNS-only monitoring described in source reporting or analyst synthesis. |

## 8. Infrastructure and Indicators

| Indicator / Infrastructure                                            | Type                      | Operational Use                                                            |
|-----------------------------------------------------------------------|---------------------------|----------------------------------------------------------------------------|
| No normalized hard IOCs available in collected public source set      | Source limitation         | Hunt on behaviors and telemetry patterns instead of relying on blocklists. |
| Cloud-hosted C2 / direct-IP / SaaS identity patterns where applicable | Behavioral infrastructure | Review egress, proxy, Entra and EDR logs for suspicious flows.             |

## 9. Detection Engineering

This section includes deployable starting points. Tune filters to local baselines before production alerting.

### 9.1. Detection Summary

| Signal                                                                        | Telemetry Source     | Analytic Goal                              |
|-------------------------------------------------------------------------------|----------------------|--------------------------------------------|
| Suspicious scripting or developer-runtime execution from downloads/temp paths | EDR process creation | Catch lure-driven malware execution.       |
| Direct-to-IP or rare ASN egress from user endpoints                           | Network/EDR          | Catch DNS-bypass C2 and unusual beaconing. |

|                                                      |                |                                                         |
|------------------------------------------------------|----------------|---------------------------------------------------------|
| Risky cloud identity sign-ins and MFA reset patterns | Entra/IdP logs | Catch credential-theft and social-engineering outcomes. |
|------------------------------------------------------|----------------|---------------------------------------------------------|

## 9.2. Sigma / Detection Content

```

title: Direct-to-IP Malware C2 and DNS Bypass Detection Priorities - Suspicious Activity Heuristic
id: auto-generated-direct-to-ip-malware-c2-and-dns-bypass-detection
status: experimental
description: Detects process or network command-line clues that should be reviewed in context for this report.
logsource:
  product: windows
  category: process_creation
detection:
  selection_cmd:
    CommandLine|contains:
      - 'Multiple malware families and intrusion operators'
      - 'Application-layer C2'
      - 'Direct IP egress'
      - 'Proxy bypass'
  condition: selection_cmd
falsepositives:
  - Administrative validation, vulnerability scanning, or approved red-team testing.
level: medium
  
```

## 9.3. Hunt Query

```

// Hunt for suspicious activity related to Direct-to-IP Malware C2 and DNS Bypass Detection Priorities
DeviceNetworkEvents
| where Timestamp > ago(14d)
| where RemoteUrl has_any ('Multiple malware families and intrusion operators', 'Application-layer C2',
'Direct IP egress', 'Proxy bypass', 'Defense evasion against DNS-only monitoring') or
InitiatingProcessCommandLine has_any ('Multiple malware families and intrusion operators',
'Application-layer C2', 'Direct IP egress', 'Proxy bypass', 'Defense evasion against DNS-only
monitoring')
| project Timestamp, DeviceName, InitiatingProcessAccountName, InitiatingProcessFileName, RemoteUrl,
RemoteIP, InitiatingProcessCommandLine
  
```

# 10. Threat Hunting

## 10.1. Hunt Hypothesis

If this activity is present, telemetry should show a cluster of suspicious user execution, rare network egress, risky sign-in behavior, or web-server process anomalies connected by account, device, or time window.

## 10.2. Hunt Query

```

let lookback = 14d;
let suspect_terms = dynamic(['Multiple malware families and intrusion operators', 'Application-layer C2',
'Direct IP egress', 'Proxy bypass', 'Defense evasion against DNS-only monitoring']);
DeviceProcessEvents
| where Timestamp > ago(lookback)
| where ProcessCommandLine has_any (suspect_terms) or FolderPath has_any ("\\Downloads\\", "\\Temp\\",
"/tmp/", "/var/www/")
| summarize FirstSeen=min(Timestamp), LastSeen=max(Timestamp), Cmds=makeset(ProcessCommandLine, 10) by
DeviceName, InitiatingProcessAccountName, FileName
| order by LastSeen desc
  
```

# 11. Risk Assessment

| Dimension  | Rating | Rationale                                                                     |
|------------|--------|-------------------------------------------------------------------------------|
| Likelihood | Medium | Current public reporting and common exposure paths make opportunistic contact |

|                         |      |                                                                                                        |
|-------------------------|------|--------------------------------------------------------------------------------------------------------|
|                         |      | plausible.                                                                                             |
| Impact                  | High | Credential theft, malware execution, or shared-platform compromise can produce operational disruption. |
| Detection Maturity Need | High | Behavioral analytics are required because hard IOCs may be absent or short-lived.                      |

## 12. Recommended Actions

| Priority | Owner                    | Action                                                                                          |
|----------|--------------------------|-------------------------------------------------------------------------------------------------|
| Critical | SOC                      | Run the included KQL hunts and review hits with identity and endpoint context.                  |
| High     | Vulnerability / Platform | Confirm public-service patching, exposed admin interfaces, and third-party access restrictions. |
| High     | Identity                 | Enforce phishing-resistant MFA for privileged and developer accounts; review risky sign-ins.    |
| Medium   | Resilience               | Update incident runbooks for third-party/shared-platform disruption.                            |

## 13. References

- [1] Unit 42 direct-to-IP malware C2: <https://unit42.paloaltonetworks.com/malware-bypass-dns-direct-to-ip/>
- [2] Unit 42 threat research index: <https://unit42.paloaltonetworks.com/>
- [3] M-Trends 2026: <https://cloud.google.com/blog/topics/threat-intelligence/m-trends-2026>